

10 (**short**) A satellite of mass m is in an elliptical orbit around Earth. At its perigee (point of closest approach) it has an altitude of 1200 km and a speed of 7900 m s^{-1} . The radius of Earth is 6400 km, the mass of Earth is $6 \times 10^{24} \text{ kg}$ and the gravitational constant G is $6.67 \times 10^{-11} \text{ m}^3 \text{ kg}^{-1} \text{ s}^{-2}$.

(a) If the work done on the satellite to put it into orbit was $1 \times 10^{11} \text{ J}$, determine the mass of the satellite. [6]

(b) If the velocity of the satellite is 4000 m s^{-1} at its apogee (point on its orbit furthest from Earth), determine its altitude at this point. [4]

11 (**short**) Figure 7 shows a mass–spring–dashpot system, where f is an applied force, m is the mass of the body, k is the spring stiffness and λ is the viscosity of the dashpot.

(a) Derive the equation of motion for this system. [5]

(b) If $f = F \sin(0.8\omega_n t)$, where F is constant, ω_n is the natural frequency and t is time, estimate the minimum required damping factor such that the amplitude of the response does not exceed $1.2F/k$. [5]

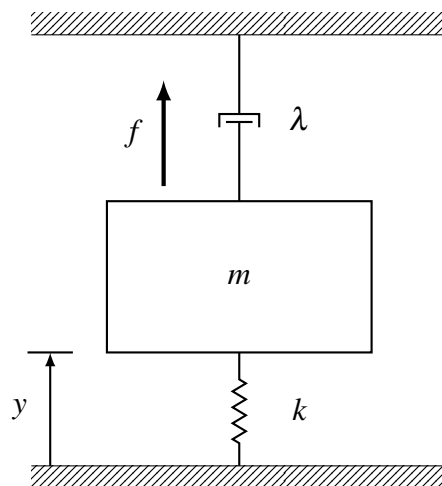


Fig. 7